

BUILDING RESEARCH PROJECTS

Exposures, outcomes, and the covariates in between

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Two hours, two halves

HOUR 1

Training · 60 min

- 0:00 Welcome & why this matters
- 0:05 Anatomy of a research question (PECO)
- 0:15 Exposure and outcome
- 0:25 The covariate family
- 0:40 Drawing your thinking — DAGs
- 0:50 Activity 1 — Build a DAG

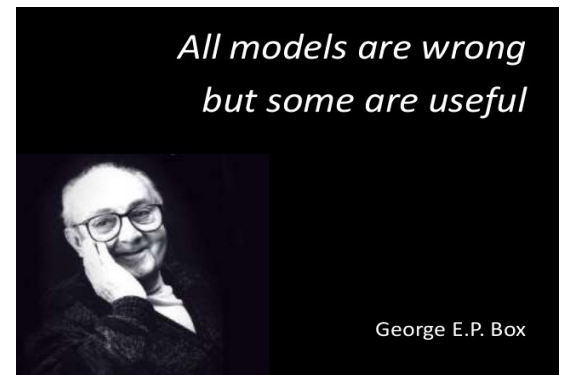
HOUR 2

Paper work · 60 min

- 1:00 The ANC question we will answer
- 1:10 Our PECO and our DAG
- 1:25 Picking how to measure each variable
- 1:35 Analysis: descriptive → crude → adjusted
- 1:50 Activity 2 — first descriptive table
- 1:58 Manuscript structure & next steps

Times are guidelines — the facilitator should adjust based on the room.

Why we slow down before we model



A model is only as good as the question it was built to answer. Researchers who run regressions without first specifying exposure, outcome, and covariates end up with results they cannot defend.



Without a clear question

...you cannot tell which result matters. Every coefficient looks equally interesting and no result is actionable.



Without a causal map

...you don't know which variables to adjust for, which to leave alone, and which to actively avoid touching.



Without that discipline

...your findings are vulnerable to reviewers and — more importantly — your conclusions may guide care badly.

Every research question has four parts

PECO/PICO — the discipline of writing your question down before you write your code.



Population

Who are we studying?

Pregnant women attending Mayuge HC IV and Buluba Hospital in 2023–24



Exposure/Intervention

What is the factor we think matters?

Late ANC1 — first visit at ≥ 14 weeks gestational age



Comparison

Compared to what?

Early ANC1 — first visit at < 14 weeks gestational age



Outcome

What do we want to predict or explain?

ANC4+ completion (yes / no)

Structure of a Research Project

A good research project includes:

- Research problem
- Research question
- Exposure(s)
- Outcome(s)
- Covariates
- Study design
- Data collection
- Statistical analysis

Exposure — the variable we think drives the outcome

WHAT IT IS
The factor whose effect we want to estimate. The ‘E’ in our research question. Often called the ‘independent variable’ or ‘risk factor’.

An **exposure** is the factor believed to influence a health outcome.

THREE KEY DECISIONS

- Continuous, categorical, or binary?
- If categorical: what is the reference group?
- Where is the cut-off — and why that one?

Research Topic	Exposure
ANC utilization	Number of ANC visits
Child nutrition	Breastfeeding practice
Maternal health	Maternal anemia
Delivery care	Skilled birth attendance

ANC EXAMPLE

Late ANC1

Defined as:

```
late_anc1 = anc_gestationalage >= 14
           at first recorded visit
```

WHY THIS CUT-OFF

WHO ANC 2016 recommends first contact before 12 weeks. We use 14 weeks because it gives enough records on both sides for a stable comparison — and it's clinically meaningful.

Outcome — the variable we want to explain

WHAT IT IS

The thing we are trying to predict or explain. The 'O' in our question. Often called the 'dependent variable' or simply the 'event'.

An **outcome** is the health event or condition being studied.

WHAT A GOOD OUTCOME LOOKS LIKE

- Clearly defined (everyone agrees what counts)
- Reliably recorded (or easily derived)
- Clinically meaningful (someone cares about it)
- Happens often enough to study

ANC EXAMPLE

ANC4+ completion

Defined as:

```
anc4_plus = number_of_visits >= 4  
            within the pregnancy window
```

WHY THIS OUTCOME

Binary, easy to derive after linkage, clinically meaningful (WHO ANC4+ indicator), and present in our data for every linked patient.

The simplest possible model

Before we add anything, every research project starts here: one exposure, one outcome, and an arrow connecting them.



Does late first ANC visit cause reduced ANC4+ completion?

But the world is messier than this. Other things influence the exposure, the outcome, or both — and we have to think about them.

Four kinds of covariate — and they behave differently

Not every variable is the same. Knowing what role a variable plays tells us whether to adjust for it, ignore it, or treat it as part of the story.

C

Confounder

Causes both exposure and outcome

ACTION: **ADJUST**

M

Mediator

On the causal path $E \rightarrow \dots \rightarrow O$

ACTION: **USUALLY NOT**

X

Effect modifier

Changes how strong $E \rightarrow O$ is

ACTION: **STRATIFY**

K

Collider

Common effect of E and O

ACTION: **DON'T TOUCH**

Confounder

DEFINITION

A variable that causes both the exposure AND the outcome — creating a spurious association we must remove.

HOW TO RECOGNISE ONE

Ask: 'Does this variable plausibly cause the exposure?' AND 'Does it plausibly cause the outcome?' If both yes — confounder.

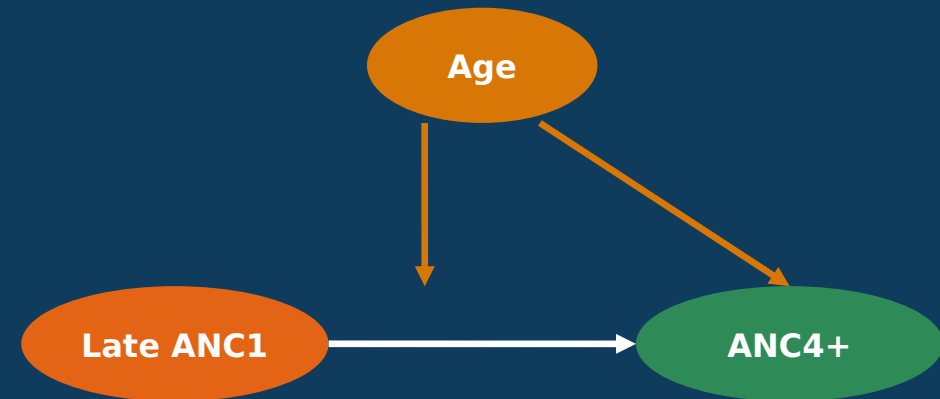
WHAT TO DO

Adjust for it. Include it in the model as a covariate; stratify if appropriate.

ANC EXAMPLE

Maternal age

Older mothers may both present later for ANC AND be more likely to complete ANC4+ for other reasons (parity, experience).



Mediator

DEFINITION

A variable on the causal pathway BETWEEN exposure and outcome — it explains HOW the exposure affects the outcome.

HOW TO RECOGNISE ONE

Ask: 'Does the exposure cause this variable, AND does this variable then cause the outcome?' If yes — mediator.

WHAT TO DO

Usually do NOT adjust. Adjusting closes the path you are trying to estimate.

ANC EXAMPLE

Receipt of antenatal package

Women who arrive late may receive fewer ANC services at first visit — and that fact in turn affects their ANC4+ likelihood.



The mediator sits ON the path.

Effect modifier (moderator)

DEFINITION

A variable that changes how STRONG the exposure-outcome relationship is for different groups.

HOW TO RECOGNISE ONE

Ask: 'Does the effect of E on O differ depending on the level of this variable?' If yes — effect modifier.

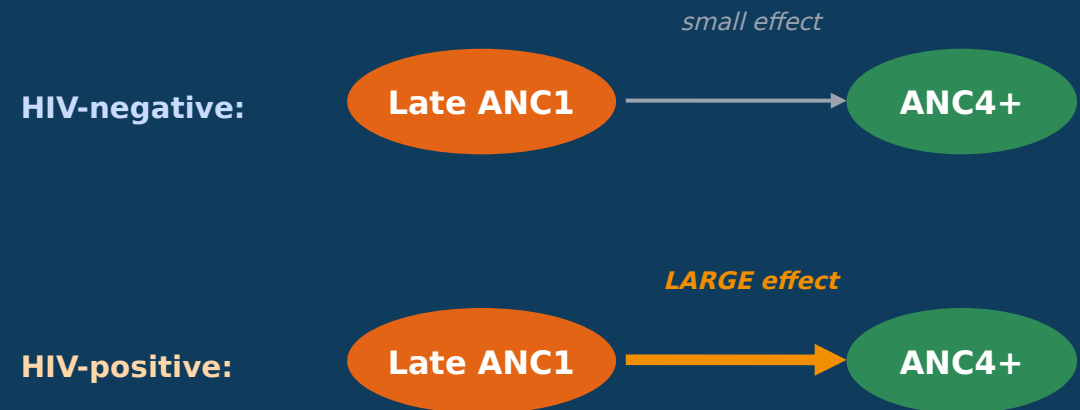
WHAT TO DO

Don't adjust away — stratify or include an interaction term in the model.

ANC EXAMPLE

HIV status

The effect of late ANC1 on ANC4+ might be larger for HIV-positive women (who need closer follow-up) than HIV-negative.



Collider — the dangerous one

DEFINITION

A variable that is CAUSED by both the exposure and the outcome — adjusting for it CREATES a spurious association.

HOW TO RECOGNISE ONE

Ask: 'Do both the exposure AND the outcome cause this variable?' If yes — collider. Often unrecorded variables.

WHAT TO DO

Do NOT adjust. Do NOT condition on it. Doing so is one of the most common analysis mistakes.

ANC EXAMPLE

Being included in the analysis dataset

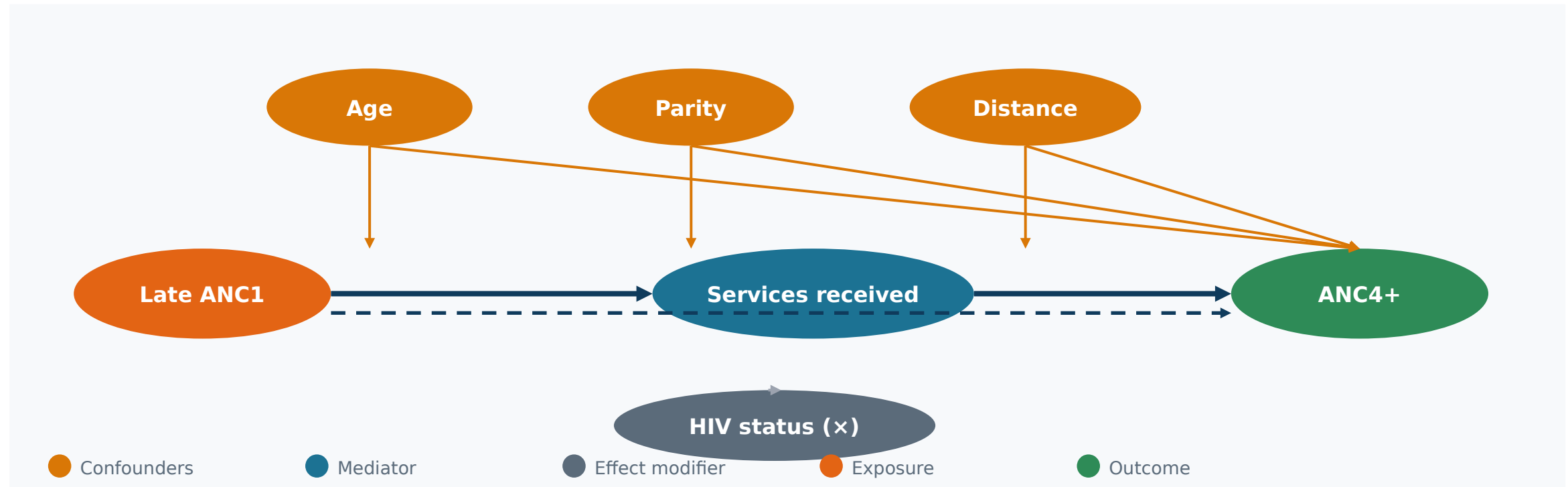
Both late ANC1 (more likely to drop out) AND non-completion (no further visits to link) reduce the chance of appearing in our final linked dataset.



If we adjust for 'in dataset', we create a fake link between E and O.

Drawing your thinking: the DAG

A directed acyclic graph (DAG) is a simple picture of your assumptions. Nodes are variables, arrows are causal relationships. Drawing one forces you to be explicit.



Build a DAG for a different question

Different research question. Same skill. Work in pairs (7 min), then we share two DAGs at the front (3 min).

THE QUESTION

Is maternal anaemia at first ANC visit associated with low birth weight?

VARIABLES TO CONSIDER

Maternal age · Parity · Maternal BMI · Iron supplementation · Gestational age at delivery · HIV status · Wealth / education

YOUR TASK

Draw a DAG. Place each variable as a node. Draw arrows showing what you think causes what. Colour-code: confounder / mediator / effect modifier / collider.

HINTS

- Start by drawing E and O alone.
- For each other variable, ask: does it cause E? does it cause O? does it sit between them?
- There are no right answers — only well-reasoned ones. Be ready to defend your arrows.
- Use paper, a whiteboard, or a phone app — whatever lets you draw fast.

END OF PART 1

Now to our paper.

We will apply this framework to a real question we can answer with our linked ANC data — and walk it from PECO to first results.

✂ We should be here at minute 60

The question we will answer

RESEARCH QUESTION

Is late first ANC visit (≥ 14 weeks gestational age) associated with failure to complete ANC4+ at Mayuge HC IV and Buluba Hospital, 2023-2024?

WHY IT MATTERS

ANC4+ is a WHO indicator. Identifying women at risk of dropout enables targeted follow-up.

WHAT WE'D NEED

Per-woman first-visit gestational age, total visit count, and basic demographics.

WHAT WE HAVE

1,759 linked patients across 2 facilities with all of the above already derived.

Our PECO, written out

P**Population**

Pregnant women attending antenatal care at Mayuge HC IV or Buluba Hospital between 1 Jan 2023 and 31 Dec 2024, who are linked to at least one identified patient record (n = 1,759).

E**Exposure**

Late first ANC visit, defined as gestational age ≥ 14 weeks at the woman's first recorded ANC contact.

C**Comparison**

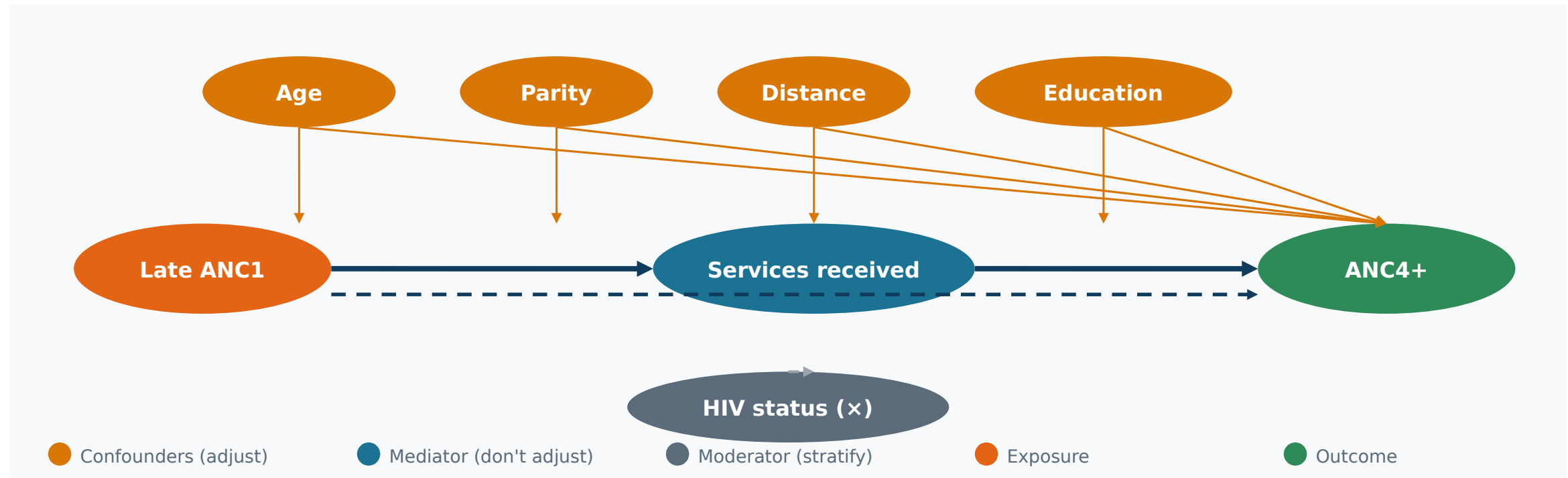
Early first ANC visit, defined as gestational age < 14 weeks at the woman's first recorded ANC contact.

O**Outcome**

Completion of ≥ 4 ANC visits (ANC4+) within the pregnancy window for that patient.

Our DAG — the causal map we will defend

We will adjust for the amber confounders. We will NOT adjust for the teal mediator (it's on the path). HIV status is treated as a moderator via stratification.



How we measure each variable

Every node in the DAG becomes a column in our dataset. Each must have a clear definition.

Variable	Role	Operational definition	Type
Late ANC1	exposure	anc_gestationalage >= 14 at first visit	Binary
ANC4+	outcome	n_visits >= 4 within pregnancy window	Binary
Maternal age	confounder	ageyrs at first visit (years)	Continuous
Parity	confounder	anc_parity at first visit	Continuous / categorical
Distance	confounder	Subcounty distance band to facility	Categorical
Education	confounder	Highest level recorded (proxy)	Categorical
Services received	mediator	Count of ANC interventions at first visit	Continuous
HIV status	modifier	emtct_codes at any visit	Binary

Three stages of analysis

Every paper of this type follows the same three stages. We build them up sequentially so reviewers can see the story.

1

Descriptive

Who is in the data?

Table 1: characteristics of the population, split by exposure (late vs early ANC1). Tells us if our two groups even look comparable.

→ Produces Table 1

2

Crude association

Does the unadjusted relationship exist?

Simple 2×2 table and a univariate logistic regression of exposure on outcome. Sets the baseline effect before any adjustment.

→ Produces Table 2

3

Adjusted model

What's the effect after adjustment?

Multivariable logistic regression adjusting for the amber confounders from our DAG. Optionally stratify by HIV status.

→ Produces Table 3 + sensitivity

Step 1 — Build Table 1 by exposure group

```
library(tidyverse); library(gtsummary)

# Derive exposure and outcome
anc_pt <- anc_pt %>%
  mutate(
    late_anc1 = anc_gestationalage_first >= 14,
    anc4_plus = n_visits >= 4
  )

# Table 1 – characteristics by exposure
anc_pt %>%
  select(late_anc1, ageyrs, anc_parity,
         distance_band, education_cat,
         hiv_status, anc4_plus) %>%
  tbl_summary(by = late_anc1,
             missing = 'ifany') %>%
  add_p()    # crude test of difference
```

WHAT WE GET

Table 1

Two columns: late ANC1 vs early. Rows show age, parity, distance, education, HIV, and the crude outcome. Tells us if the groups are balanced before any modelling.

Step 2 — Crude association

```
# 2x2 table
anc_pt %>%
  count(late_anc1, anc4_plus) %>%
  pivot_wider(names_from = anc4_plus,
              values_from = n)

# Crude odds ratio with 95% CI
crude <- glm(anc4_plus ~ late_anc1,
            data = anc_pt,
            family = binomial)

broom::tidy(crude, exponentiate = TRUE,
            conf.int = TRUE)
```

WHAT WE GET

Table 2

An unadjusted odds ratio with 95% CI for the effect of late ANC1 on ANC4+ completion. This is the headline number that the adjusted model will then refine.

Step 3 — Adjusted model

```
# Adjust for the amber confounders only
# (NOT services_received — that's a mediator)

adjusted <- glm(
  anc4_plus ~ late_anc1 + ageyrs +
               anc_parity + distance_band +
               education_cat,
  data = anc_pt,
  family = binomial
)

tbl_regression(adjusted, exponentiate = TRUE)

# Sensitivity: stratify by HIV status
anc_pt %>% split(.$hiv_status) %>%
  map(~ glm(anc4_plus ~ late_anc1 + ageyrs +
             anc_parity + distance_band +
             education_cat,
             data = ., family = binomial))
```

WHAT WE GET

Table 3

Adjusted odds ratios for late ANC1 and each confounder. The crude vs adjusted comparison shows how much the relationship is explained by confounding.

Plus stratified results by HIV status (effect modifier).

Build the first descriptive table — together

Three tasks on the linked ANC dataset. The facilitator runs the code; you follow on your machine and report back.

Task A5 min	Task B5 min	Task C5 min
Define exposure & outcome WHAT Compute late_anc1 (≥ 14 wks GA at first visit) and anc4_plus (≥ 4 visits). REPORT BACK How many women in each exposure group? What's the crude % completing ANC4+?	Table 1 WHAT Use tbl_summary(by = late_anc1) on age, parity, distance, education, HIV. REPORT BACK Which characteristic differs most between the two exposure groups? Why might that matter?	Crude OR WHAT Fit glm(anc4_plus ~ late_anc1) and exponentiate the coefficient. REPORT BACK What's the crude odds ratio and its 95% CI? Statistically significant?

What we walk out with

TABLES

- T1 — Characteristics of patients by ANC1 timing group
- T2 — Crude association between late ANC1 and ANC4+
- T3 — Adjusted OR with 95% CI for late ANC1 on ANC4+
- T4 — Stratified results by HIV status (effect modification)

FIGURES

- F1 — Forest plot of crude vs adjusted OR
- F2 — Stratified estimates by HIV status
- F3 — DAG figure as presented in the methods
- F4 — Sensitivity analysis: varying the cut-off

Manuscript outline

Introduction

Why ANC4+ matters; what we know about late-presentation predictors; the gap in LMIC settings.

Methods

Setting, dataset (linked cohort), PECO, DAG-justified covariate selection, model specification, sensitivity.

Results

Population characteristics, crude OR, adjusted OR, stratified by HIV status, sensitivity to the 14-week cut-off.

Discussion

What the effect size means clinically, comparison with regional studies, limitations (especially the collider note).

Conclusion

Implications for ANC programmes; women presenting late are flagged for active follow-up.

Three things to take home



Write the question first

PECO before code. If you cannot write your research question in one sentence, you are not ready to model.



Draw the DAG

Every variable in your dataset has a role. The DAG forces you to name it before you decide what to adjust for.



Adjust for confounders only

Don't adjust for mediators (closes your effect). Don't condition on colliders (creates fake associations).

FROM QUESTION TO PAPER

We have a question, a DAG, and a model.

Now the work is execution: fit the models, write the methods, draft the results, send the paper out. The framework is in place — the rest is craft.

Questions, comments, who-does-what